Unit - 2

3 a] If R and S are fuzzy relations given by

$$R = \begin{array}{c} \\ x_1 \\ x_2 \\ x_3 \end{array} \begin{array}{c} \begin{array}{cc} y_1 & y_2 \end{array} \\ \begin{bmatrix} 0.5 & 0.1 \\ 0.2 & 0.9 \\ 0.8 & 0.6 \end{bmatrix} \end{array}$$

and

$$S = \begin{array}{c} \\ x_1 \\ x_2 \\ x_3 \end{array} \begin{array}{c} \begin{array}{cc} y_1 & y_2 \end{array} \\ \begin{bmatrix} 0.6 & 0.5 \\ 0.4 & 0.8 \\ 0.7 & 0.9 \end{bmatrix} \end{array}$$

Find $R \cup S$ and $R \cap S$.

→

$R \cap S = \min(\mu_R, \mu_S)$

$R \cup S = \max(\mu_R, \mu_S)$

Now from matrix table.

$$R = \left\{ \frac{0.5}{x_1y_1} \quad \frac{0.1}{x_1y_2} \quad \frac{0.2}{x_2y_1} \quad \frac{0.9}{x_2y_2} \quad \frac{0.8}{x_3y_1} \quad \frac{0.9}{x_3y_2} \right\}$$

$$S = \left\{ \frac{0.6}{x_1y_1} \quad \frac{0.5}{x_1y_2} \quad \frac{0.4}{x_2y_1} \quad \frac{0.8}{x_2y_2} \quad \frac{0.7}{x_3y_1} \quad \frac{0.9}{x_3y_2} \right\}$$

Now

$$R \cap S = \left\{ \frac{0.5}{x_1y_1} \quad \frac{0.1}{x_1y_2} \quad \frac{0.2}{x_2y_1} \quad \frac{0.8}{x_2y_2} \quad \frac{0.7}{x_3y_1} \quad \frac{0.9}{x_3y_2} \right\}$$

$$R \cap S = \begin{array}{c} \\ x_1 \\ x_2 \\ x_3 \end{array} \begin{array}{c} \begin{array}{cc} y_1 & y_2 \end{array} \\ \begin{bmatrix} 0.5 & 0.1 \\ 0.2 & 0.8 \\ 0.7 & 0.9 \end{bmatrix} \end{array}$$

$$R \cup S = \left\{ \begin{matrix} 0.6 & 0.5 & 0.4 & 0.9 & 0.8 & 0.9 \\ x_1y_1 & x_1y_2 & x_2y_1 & x_2y_2 & x_3y_1 & x_3y_2 \end{matrix} \right\}$$

$$\begin{bmatrix} 0.6 & 0.5 \\ 0.4 & 0.9 \\ 0.8 & 0.9 \end{bmatrix}$$

$$\bar{R} = 1 - (\text{~~R~~} \; RRS)$$

$$= \left\{ \begin{matrix} 0.5 & 0.9 & 0.8 & 0.1 & 0.2 & 0.4 \\ x_1y_1 & x_1y_2 & x_2y_1 & x_2y_2 & x_3y_1 & x_3y_2 \end{matrix} \right\}$$

$$\begin{array}{c} \\ x_1 \\ x_2 \\ x_3 \end{array} \begin{array}{c} \begin{matrix} y_1 & \quad y_2 \end{matrix} \\ \begin{bmatrix} 0.5 & 0.9 \\ 0.8 & 0.1 \\ 0.2 & 0.4 \end{bmatrix} \end{array}$$

b) If Fuzzy sets are $A = \frac{0.3}{30} + \frac{0.7}{60} + \frac{1}{100}$ $B = \frac{0.2}{20} + \frac{0.4}{40} + \frac{0.6}{60}$

$C = \frac{0.3}{50} + \frac{0.6}{100}$, Find Fuzzy relation $R = A \times B$. $S = B \times C$

Also find $R = S$

→

We have $\mu_R = A \times B = \min(\mu_A, \mu_B)$

& $\mu_S = B \times C = \min(\mu_B, \mu_C)$

$$R = \begin{matrix} \\ 30 \\ 60 \\ 100 \end{matrix} \begin{matrix} \begin{matrix} 20 & 40 & 60 \end{matrix} \\ \begin{bmatrix} 0.2 & 0.3 & 0.3 \\ 0.2 & 0.4 & 0.6 \\ 0.2 & 0.4 & 0.6 \end{bmatrix} \end{matrix} \qquad S = \begin{matrix} \\ 20 \\ 40 \\ 60 \end{matrix} \begin{matrix} \begin{matrix} 50 & 100 \end{matrix} \\ \begin{bmatrix} 0.2 & 0.2 \\ 0.3 & 0.4 \\ 0.3 & 0.6 \end{bmatrix} \end{matrix}$$

Now

~~R~~ $= R = S$

We have

$(30, 50) = \max\{\min(0.2, 0.2) \ \min(0.3, 0.3) \ \min(0.3, 0.3)\}$

$(0.2, 0.3, 0.3) = 0.3$

$(30, 100) = \max\{\min(0.2, 0.2) \ \min(0.3, 0.4) \ \min(0.3, 0.6)\}$

$(0.2, 0.3, 0.3) = 0.3$

$(60, 50) = \max\{\min(0.2, 0.2) \ \min(0.4, 0.3) \ \min(0.6, 0.3)\}$

$(0.2, 0.3, 0.3) = 0.3$

$(60, 100) = \max\{\min(0.2, 0.2) \ \min(0.4, 0.4) \ \min(0.6, 0.6)\}$

$(0.2, 0.4, 0.6) = 0.6$

$R = S(100, 100) = \max\{\min\{0.2, 0.2\}, \min\{0.4, 0.4\}, \min\{0.6, 0.6\}\}$

$= \max\{0.2, 0.4, 0.6\} = 0.6$

$\therefore T = R \circ S$

	50	100
30	0.3	0.3
60	0.3	0.6
100	0.3	0.6

Q.4] For the fuzzy sets A and B define A - B and $A \oplus B$

If $A = \frac{0.2}{x1} + \frac{0.5}{x2} + \frac{0.6}{x3}$ and $\frac{0.1}{x1} + \frac{0.4}{x2} + \frac{0.5}{x3}$ then find A - B and $A \oplus B$.

→

$A - B = \max[0 (\mu A(x) - \mu B(x))]$

$A \oplus B = \min[1 (\mu A(x) + \mu B(x))]$

$A - B = \max [0 (\mu A(x) - \mu B(x))]$

$[0 \cdot (0.2 - 0.1)]$

0.1

$\max [0 \cdot 0.5 - 0.4]$

0.1

$\max [0, 0.6 - 0.5]$

0.1

$A - B = \frac{0.1}{x1} + \frac{0.1}{x2} + \frac{0.1}{x3}$

$\min [1, (\mu_A(x) + \mu_B(x))]$

$\min [1, 0.2 + 0.1]$

0.3

$\min [1, 0.5 + 0.4]$

0.9

$\min [1, 0.6 + 0.5]$

$[1, 1]$

$A \oplus B = \frac{0.3}{x_1} + \frac{0.9}{x_2} + \frac{1}{x_3}$

If $A = \left\{ \frac{0.3}{30}, \frac{0.7}{60}, \frac{1}{100}, \frac{0.2}{120} \right\}$ and $B = \left\{ \frac{0.2}{20}, \frac{0.4}{40}, \frac{0.6}{60}, \frac{0.8}{80} \right\}$ Are Fuzzy sets then find the relation $R = A \times B$ Also R

→

We have $R = A \times B$ and

→ $\mu_{A \times B}(x, y) = \min \{ \mu_A(x), \mu_B(y) \}$

$R = \{ \frac{0.2}{30,20}, \frac{0.3}{30,40}, \frac{0.3}{30,60}, \frac{0.3}{30,80}, \frac{0.2}{60,20}, \frac{0.4}{60,40}, \frac{0.6}{60,60}, \frac{0.7}{60,} \rightarrow$

$\frac{0.2}{100,20}, \frac{0.4}{100,40}, \frac{0.6}{100,60}, \frac{0.8}{100,80}, \frac{0.2}{120,20}, \frac{0.2}{120,40}, \frac{0.2}{120,60}, \frac{0.2}{120,} \rightarrow$

Now In matrix from

$$R = MR = \begin{array}{c} \\ 30 \\ 60 \\ 100 \\ 120 \end{array} \begin{bmatrix} 20 & 40 & 60 & 80 \\ 0.2 & 0.3 & 0.3 & 0.3 \\ 0.2 & 0.4 & 0.6 & 0.7 \\ 0.2 & 0.4 & 0.6 & 0.8 \\ 0.2 & 0.2 & 0.2 & 0.2 \end{bmatrix}$$

Now $R^2 = R + R$

$$\begin{array}{c} \\ 30 \\ 60 \\ 100 \\ 120 \end{array} \begin{bmatrix} 20 & 40 & 60 & 80 \\ 0.2 & 0.3 & 0.3 & 0.3 \\ 0.2 & 0.4 & 0.6 & 0.7 \\ 0.2 & 0.4 & 0.6 & 0.8 \\ 0.2 & 0.2 & 0.2 & 0.2 \end{bmatrix}$$

If R and S are Fuzzy relations given by

$$R = \begin{array}{c} \\ x_1 \\ x_2 \\ x_3 \end{array} \begin{array}{c} \begin{array}{cc} y_1 & y_2 \end{array} \\ \begin{bmatrix} 0.5 & 0.1 \\ 0.2 & 0.9 \\ 0.8 & 0.6 \end{bmatrix} \end{array} \qquad S = \begin{array}{c} \\ x_1 \\ x_2 \\ x_3 \end{array} \begin{bmatrix} 0.6 & 0.5 \\ 0.4 & 0.8 \\ 0.7 & 0.9 \end{bmatrix}$$

Then Find R ∪ S R ∩ S

$$R = \left\{ \frac{0.5}{x_1y_1} \quad \frac{0.1}{x_1y_2} \quad \frac{0.2}{x_2y_1} \quad \frac{0.9}{x_2y_2} \quad \frac{0.8}{x_3y_1} \quad \frac{0.6}{x_3y_2} \right\}$$

$$S = \left\{ \frac{0.6}{x_1y_1} \quad \frac{0.5}{x_1y_2} \quad \frac{0.4}{x_2y_1} \quad \frac{0.8}{x_2y_2} \quad \frac{0.7}{x_3y_1} \quad \frac{0.9}{x_3y_2} \right\}$$

Now $R \cap S = \min(\mu_A, \mu_B)$

$$R \cap S = \frac{0.5}{x_1y_1} \quad \frac{0.1}{x_1y_2} \quad \frac{0.2}{x_2y_1} \quad \frac{0.8}{x_2y_2} \quad \frac{0.7}{x_3y_1} \quad \frac{0.9}{x_3y_2}$$

$$\begin{bmatrix} 0.5 & 0.1 \\ 0.2 & 0.8 \\ 0.7 & 0.9 \end{bmatrix}$$

$$R \cup S = \frac{0.6}{x_1y_1} \quad \frac{0.5}{x_1y_2} \quad \frac{0.4}{x_2y_1} \quad \frac{0.9}{x_2y_2} \quad \frac{0.8}{x_3y_1} \quad \frac{0.9}{x_3y_2}$$

$$\begin{bmatrix} 0.5 & 0.1 \\ 0.2 & 0.8 \\ 0.7 & 0.9 \end{bmatrix}$$

4 Q) Consider the fuzzy sets A, B, C defined on the interval $x = [0, 10]$ of integers by the membership grade function $\mu_A(x) = \frac{x}{x+2}$,

$\mu_B(x) = 2^{-x}$, $\mu_C(x) = \frac{1}{1+10(x-2)^2}$

Determine the mathematical formula for.

N	$\mu_A(x) = \frac{x}{x+2}$	$\mu_B(x) = 2^{-x}$	$\mu_C(x) = \frac{1}{1+10(x-2)^2}$
0	0	1	0.024
1	0.33	0.5	0.09
2	0.5	0.25	1
3	0.6	0.125	0.09
4	0.66	0.625	0.243
5	0.71	0.312	0.0189
6	0.75	0.0156	0.0062
7	0.77	0.0078	0.0039
8	0.8	0.0039	0.0027
9	0.818	0.0019	0.002
10	0.833	0.0009	0.0015

Here $A = \left\{ \frac{0}{0} \quad \frac{0.33}{1} \quad \frac{0.5}{2} \quad \frac{0.6}{3} \quad \frac{0.66}{4} \quad \frac{0.71}{5} \quad \frac{0.75}{6} \quad \frac{0.77}{7} \quad \frac{0.8}{8} \quad \frac{0.818}{9} \quad \frac{0.833}{10} \right\}$

$\overline{A} = 1 - A = \left\{ \frac{1}{0} \quad \frac{0.67}{1} \quad \frac{0.5}{2} \quad \frac{0.4}{3} \quad \frac{0.34}{4} \quad \frac{0.29}{5} \quad \frac{0.25}{6} \quad \frac{0.23}{7} \quad \frac{0.2}{8} \quad , \quad \frac{0.182}{9} \quad , \quad \frac{0.167}{10} \right\}$

$1 - \mu_A(x)$

If $A \cup B \cup C = \left\{ \frac{1}{0} \quad \frac{0.5}{1} \quad \frac{1}{2} \quad \frac{0.6}{3} \quad \frac{0.66}{4}, \quad \frac{0.71}{5} \right.$

$\left. \frac{0.75}{6} \quad \frac{0.77}{7}, \quad \frac{0.8}{8} \quad \frac{0.818}{9} \quad \frac{0.833}{10} \right\}$

1) Find $A \cap B$ $A \cup B$ and $A \times B$ where.

$A = \frac{0.9}{1} + \frac{0.7}{3} + \frac{0.2}{4} + \frac{0.3}{6}$

$B = \frac{0.1}{1} + \frac{0.4}{3} + \frac{0.5}{4} + \frac{0.8}{5}$

Are defined on the universe $V = \{1, 2, 3, 4, 5, 6\}$.

⟶ We know that

given

$A = \frac{0.9}{1} + \frac{0.7}{3} + \frac{0.2}{4} + \frac{0.3}{6}$ $\quad B = \frac{0.1}{1} + \frac{0.4}{3} + \frac{0.5}{4} + \frac{0.8}{5}$

$A \cap B = \min \{\mu_A(x), \mu_B(x)\}$

$\frac{0.1}{1} + \frac{0.4}{3} + \frac{0.2}{4} + \frac{0.3}{6}$

$A \cup B = \max [\mu_A + \mu_B]$

$A \cup B = \frac{0.9}{1} + \frac{0.7}{3} + \frac{0.5}{4} + \frac{0.8}{5}$

$A \times B \quad A = \frac{0.9}{1} + \frac{0.7}{3} + \frac{0.5}{4} + \frac{0.8}{5} \quad B = \frac{0.1}{2} + \frac{0.4}{3} + \frac{0.2}{4} + \frac{0.3}{6}$

c) Find α - cuts of the fuzzy set.

$$A = \frac{0.4}{v} + \frac{0.2}{w} + \frac{0.5}{x} + \frac{0.4}{y} + \frac{1}{z} \quad \text{for } \alpha = 0.2, 0.5, 0.9 \text{ and } 1$$

$\longrightarrow$

An α cut of fuzzy set A is a crisp set A_α that contains all the elements of the universal set X that have a membership grade in A grater than or equal to the specified value of α.

i.e $A_\alpha = \{x \in X \mid \mu_A(x) \geq \alpha\}$

Given $A = \frac{0.4}{v} + \frac{0.2}{w} + \frac{0.5}{x} + \frac{0.4}{y} + \frac{1}{z}$

$\therefore A_\alpha$ at $\alpha = 0.2$ will be a set of all x such that $\mu_A(x) \geq 0.2$

$A_{0.2} = \{0.4, 0.2, 0.5, 0.4, 1\}$

$= \{v, w, x, y, z\}$

A_1 $\alpha = 0.5$ we have.

$A_{0.5} = \{0.5\}$ we have.

$A_{0.5} = \{0.5, 1\} = \{x, z\}$

At $\alpha = 0.9$

$A_{0.9} = \{1\} = \{z\}$

At $\alpha = 1$

$A_1 = 1 = \{z\}$ .... Ans.

Q) Given a fuzzy set $A = \frac{0}{x_1} + \frac{0.6}{x_2} + \frac{0.5}{x_3}$

Find the measure of fuzziness of a set A.

→

Find the bounded sum for the fuzzy sets.

$A = \frac{0.5}{4} + \frac{1}{5} + \frac{0.6}{7}$ & $B = \frac{0.3}{4} + \frac{0.6}{5}$.

→ Given $A = \frac{0}{x_1} + \frac{0.6}{x_2} + \frac{0.5}{x_3}$

We have measure of fuzziness

$$F_2(A) = \frac{\text{card}(A \cdot \cap \overline{A})}{\text{cord}(A \cup \overline{A})}$$

Now

$$\overline{A} = \frac{1}{x_1} + \frac{0.4}{x_2} + \frac{0.5}{x_3}$$

$A \cap \overline{A} = \min \{\mu_A(x), \mu_A(x)\}$

$$\frac{0}{x_1} + \frac{0.4}{x_2} + \frac{0.5}{x_3}$$

$\text{cord}(A \cap \overline{A}) = 0 + 0.4 + 0.5 = 0.9$

Also $A \cup \overline{A} = \max \{\mu_A(x), \mu_A(x)\}$

$$\frac{1}{x_1} + \frac{0.6}{x_2} + \frac{0.5}{x_3}$$

$\text{cord}(A \cup \overline{A}) = 1 + 0.6 + 0.5 = 2.1$

(1) $\Rightarrow F_2(A) = \frac{0.9}{2.1} = 0.4286$

Q3 a) ——— 2

i) Equality of two fuzzy sets.

ii) Union two fuzzy sets :-

The union of fuzzy sets A and B corresponds to the connective "or". Thus $A \cup B$ = A or B The operation of complementation corresponds to negation not

iii) Difference of two fuzzy sets :- This operation is denoted by $(A - B)$. $A - B$ is the set of only those elements which exists only in set A but not in set B.

iv) Algebraic sum of two fuzzy sets.

v) Algebraic Product of two fuzzy sets.

Q3 b)
i) Fuzzy set.

Fuzzy set is a set whose elements have degrees of membership. Fuzzy sets are an extension of the classical notion of sets.

$\mu_A : A \longrightarrow [0, 1]$.